Mini Research Proposal – Applied Causal Inference Project

Deadline: June 5th 2025

Length: 2–3 pages (excluding references and figures)

Format: Docx / Odt

Each group is expected to submit a short research proposal outlining the plan for your applied causal inference project. The proposal should include the following components:

1. Title and Group Members

Example:
'The Causal Impact of Air Pollution on Hospital Admissions'
Group Members: Alice Smith (123456), Bob Lee (234567)

2. Motivation and Research Question

Example:
'Air pollution is known to correlate with various health outcomes, but establishing causality remains challenging. In this project, we aim to investigate whether increased levels of PM2.5 causally affect daily hospital admission rates in a large metropolitan area.'

3. Data Description

Example:
'We will use data from the U.S. EPA and public health records from the CDC, spanning 2010–2020 with daily resolution. Variables include PM2.5, temperature, humidity, and daily hospital admissions. Some preprocessing is needed to align time frames and handle missing data…. The data dimensions are… there are 4% missing values…'

4. Causal Assumptions

Example:
'We assume that the Causal Markov Condition and faithfulness hold. We assume no unmeasured confounders and causal stationarity…. Further assumptions about the linear or nonlinear dependency and further hyper parameter choices will be tested…'

5. Methodological Approach

Example:
Causal Discovery: We plan to use the PCMCI algorithm due to the time-series nature of our data and its ability to handle autocorrelation. Independence tests will be based on initial tests…
Causal Effect Estimation: We will estimate the causal effect using regression with optimal adjustment sets identified from the graph learned by PCMCI, and cross-check with Wright's path method….

6. Anticipated Challenges and Backup Plans

Example:

If PCMCI yields unstable graphs due to limited sample size, we will simplify our model or use domain knowledge to restrict the graph structure….

7. Toy Model Experiments

Example:

We will simulate a time series with variables X (pollution), Z (weather), and Y (hospital admissions) where $X \to Y$, $Z \to X$, and $Z \to Y$. This will help validate our pipeline and evaluate false/true positive rates of discovery methods. In the toy model we simulate the above challenges that exist in our real data…

8. Timeline and Milestones

Example:

Week 1: Finalize dataset and perform preprocessing

Week 2: Run toy model experiments

Week 3–4: Apply causal discovery and effect estimation

Week 5: Prepare final presentation

9. References

Example:

Runge, J. et al. (2019). Inferring causation from time series in Earth system sciences. Nature Communications.

Pearl, J. (2009). Causality. Cambridge University Press.